

Forder — Book IV

Proposition 17

Formal Reconstruction — Technical Documentation

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If the angle between a line AT and chord AX is congruent to the angle XYA , with Y, T on opposite sides of AX , then the circle through X, A, Y is tangent to AT at A .

1 Introduction

Forder IV.17 is the converse tangent–chord theorem.

Let X, A, Y be points of one circle, let AT be a candidate line through the circle point A , and assume that T and Y lie on opposite sides of the chord AX . If

$$\angle XAT \cong \angle XYA,$$

then the line AT is tangent to the circle at A [1].

The theorem is a converse. The angle relation itself does not assume tangency; it is used to exclude every possible secant configuration.

The formal proof is therefore naturally organized by the circle-line dichotomy:

second circle intersection or tangent.

If the tangent alternative occurs, the theorem is finished. If there is a second circle point S on the line AT , the proof must show that every possible order of S relative to the ray AT is impossible.

That produces the two characteristic branches of IV.17:

1. S lies on the same ray from A as T ;
2. S lies on the opposite ray, equivalently $S - A - T$.

The first branch is contradicted through the *opposite-side* half of IV.16 and a supplement argument. The second is contradicted through the *same-side* half of IV.16 and the exterior-angle theorem.

The production implementation is

with public endpoint

hilbert_IV17_tangent_converse

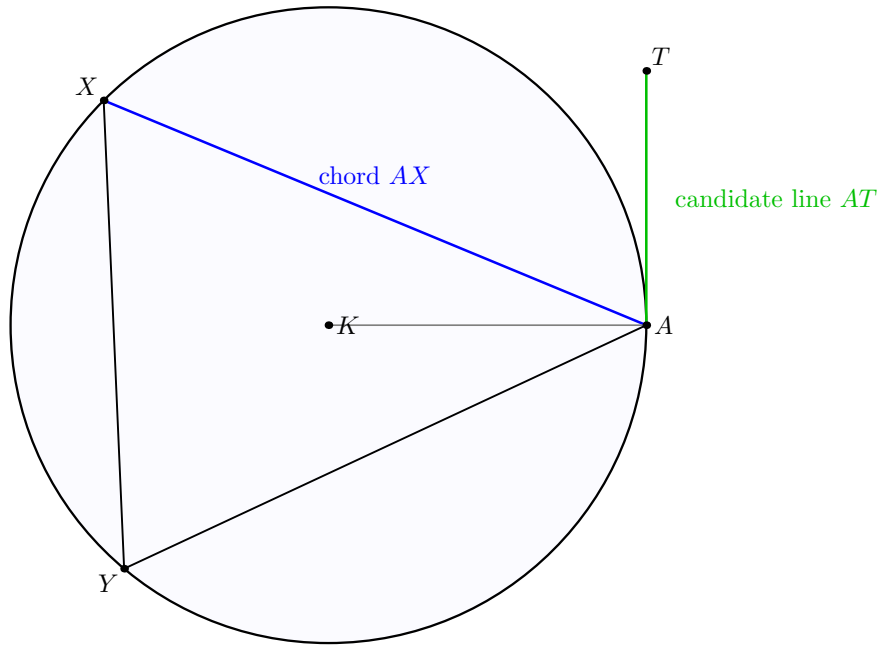
2 Classical Configuration

The basic configuration consists of chord AX , the circle point Y , and a candidate line AT through A . The side condition is

$$T \mid Y$$

relative to the chord AX .

$$\angle XAT \cong \angle XYA$$



T, Y opposite sides of AX

Figure 1: Forder IV.17. The line AT is a candidate tangent at A . The point T lies on the opposite side of chord AX from the circle point Y , and the hypothesis is $\angle XAT \cong \angle XYA$.

The theorem says that this angle condition forces AT to be the tangent carrier.

The proof does not construct the tangent independently and then compare lines. Instead, it proves that AT cannot be a secant.

3 Forder's Statement and Proof

Forder states, in substance:

If $\angle XAT = \angle XYA$ and Y, T are on opposite sides of AX , then the circle through X, A, Y touches AT at A .

His proof is by contradiction. If AT is not tangent, it meets the circle again at a point $S \neq A$. Then IV.16 determines the relation between

$$\angle XSA$$

and

$$\angle XYA$$

according to whether S and Y lie on the same or opposite sides of the chord AX .

The order of S relative to T changes which side relation holds.

In one case $\angle XSA$ becomes supplementary to $\angle XYA$, while the hypothesis and the collinearity of A, S, T identify $\angle XYA$ with the angle at A . This forces an interior triangle angle to be congruent to its own supplement, contradicting the triangle angle inequality.

In the other case $\angle XSA \cong \angle XYA$, and the hypothesis gives

$$\angle XSA \cong \angle XAT.$$

But $\angle XAT$ is an exterior angle of triangle XSA , again contradicting the exterior-angle theorem.

Thus the secant alternative is impossible, so AT is tangent.

4 Formal Interface

The production predicate is:

```
def HilbertForderIV17TangentConverse
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H] : Prop :=
  forall K R X A Y T : Geo.Point,
    forall chord tangent : Geo.Line,
      Ne X A ->
      H.OnLine X chord ->
      H.OnLine A chord ->
      H.OnLine A tangent ->
      H.OnLine T tangent ->
      Ne T A ->
      HilbertCircle Geo K R X ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R Y ->
      HilbertOppositeSide Geo T Y chord ->
      Geo.AngleCongruent X A T X Y A ->
      HilbertCircleTangentAt
        (Geo := Geo) K A tangent
```

Several design choices are visible in the interface.

First, the chord AX and the candidate tangent carrier are explicit `Geo.Line` objects.

Second, the side condition is not encoded by an orientation convention but by

```
HilbertOppositeSide Geo T Y chord
```

Third, the conclusion uses the project's tangent predicate

```
HilbertCircleTangentAt (Geo := Geo) K A tangent
```

rather than a numerical slope or perpendicular equation.

The public theorem is:

```
theorem hilbert_IV17_tangent_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV17TangentConverse (Geo := Geo)
```

5 The Tangency Predicate

The relevant interface definition is

```
def HilbertCircleTangentAt
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H]
  (K A : Geo.Point)
  (line : Geo.Line) : Prop :=
  exists T : Geo.Point,
    H.OnLine T line /\
    Ne T A /\
    Not (PrimCollinear Geo T A K) /\
    HilbertRightAngle Geo T A K
```

Thus a tangent at A is represented synthetically by a carrier through A containing a point whose ray makes a right angle with the radius AK .

IV.17 does not prove tangency by directly constructing such a perpendicular. Instead, the general circle-line dichotomy eventually returns this tangent predicate after all secant alternatives have been eliminated.

6 The Circle-Line Dichotomy

The reusable theorem

```
hilbert_circle_line_second_or_tangent
```

says that a line through a noncentral circle point A is either:

1. a secant, containing a second circle point $S \neq A$, or

2. tangent at A .

The top-level IV.17 proof begins by showing

$$K \neq A.$$

This follows because A and X are distinct circle points with equal radii from K ; if $K = A$, the segment AX would be congruent to the null segment AA , forcing $A = X$.

The dichotomy can then be applied to the candidate carrier AT .

If it returns tangency, that is exactly the theorem's conclusion.

The rest of IV.17 is therefore a proof that the secant alternative is impossible.

7 Normalizing the Secant Order

Suppose the candidate line contains a second circle point $S \neq A$.

There are three superficial possibilities along the carrier: $S = T$, S, T distinct on the same ray from A , or S on the opposite ray.

The equality case $S = T$ is absorbed into the same-ray branch through reflexivity of the same-ray relation.

For distinct S, T , the reusable production theorem

```
hilbert_secant_ray_order_split
```

normalizes the order into exactly two alternatives:

$$\text{SameRay}(A, S, T)$$

or

$$T - A - S.$$

The second relation is immediately reversed to

$$S - A - T$$

for the exterior-angle argument.

This theorem is architecturally important. During development it carried an IV.19 stage name because it was first isolated there. In the production split it was promoted to the semantic, proposition-independent name `hilbert_secant_ray_order_split` in `HilbertInterfaceIV`.

Without that promotion, IV.17 would appear to depend on the later IV.19, creating an artificial cycle. The production architecture correctly records the theorem as general line/ray order infrastructure.

8 Same-Ray Secant Branch

Assume

$$\text{SameRay}(A, S, T).$$

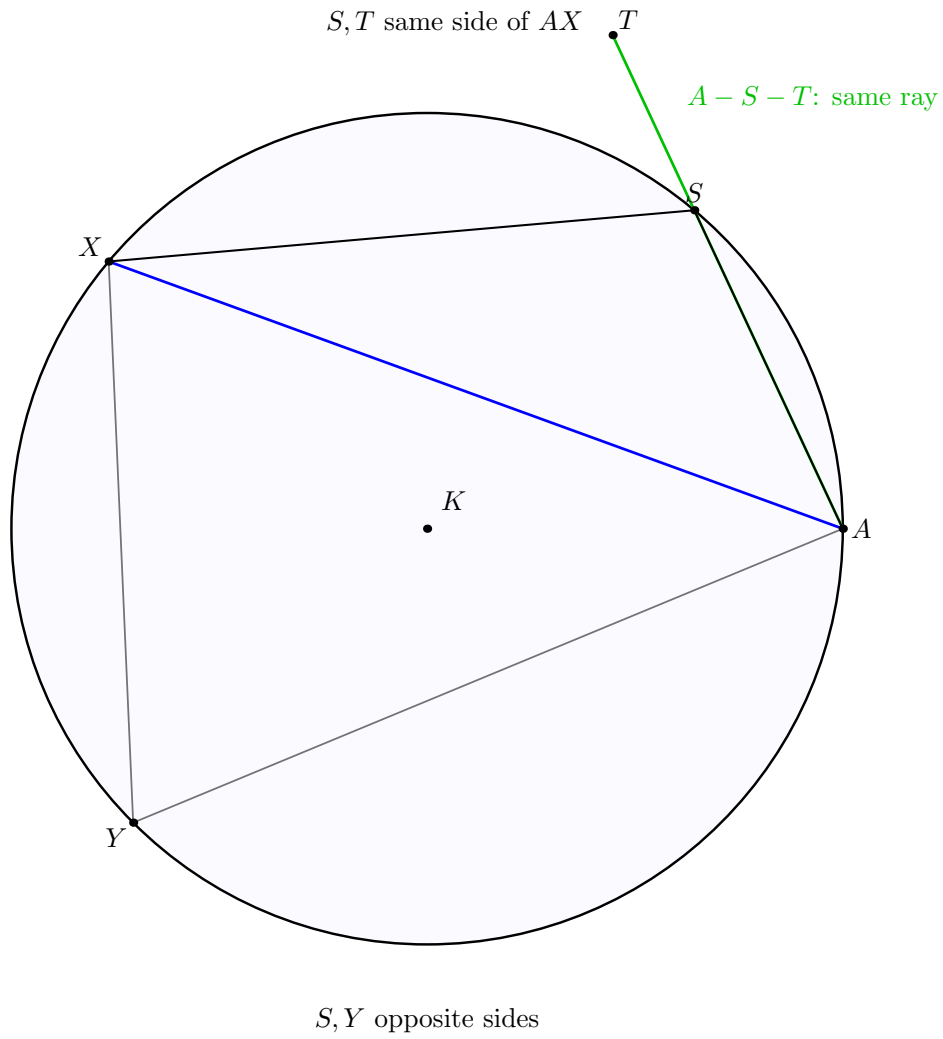


Figure 2: The impossible same-ray secant branch. The second circle intersection S and the point T lie on the same ray from A . Consequently S, T are on the same side of chord AX , while T, Y are opposite; hence S, Y are opposite.

Because S, T lie on the same ray from the chord point A , they lie on the same side of chord AX .

The theorem hypothesis says that T, Y lie on opposite sides of the chord. Transporting the side relation therefore gives

$$S \mid Y.$$

Now apply the opposite-side half of IV.16 to the circle points

$$X, A, S, Y.$$

It constructs a point U such that

$$\angle XSA \cong \angle UYA$$

and $\angle UYA$ is a supplement of

$$\angle XYA.$$

Meanwhile, because AS and AT are the same ray,

$$\angle XAS = \angle XAT.$$

The hypothesis of IV.17 gives

$$\angle XAT \cong \angle XYA,$$

hence

$$\angle XAS \cong \angle XYA.$$

After reversing the first angle:

$$\angle SAX \cong \angle XYA.$$

9 Transporting the Supplement in the Same-Ray Branch

Extend SA through A to a point V :

$$S - A - V.$$

Then $\angle XAV$ is a supplement of $\angle SAX$.

From IV.16, $\angle UYA$ is a supplement of $\angle XYA$.

Since

$$\angle SAX \cong \angle XYA,$$

Book Zero supplement transport gives

$$\angle XAV \cong \angle UYA.$$

But IV.16 also gave

$$\angle XSA \cong \angle UYA.$$

Therefore

$$\angle XSA \cong \angle XAV.$$

Now $S - A - V$, so $\angle XAV$ is the exterior angle at A of the proper triangle XSA .

The formal contradiction is supplied by the reusable theorem now owned by the interface:

hilbert_triangle_angle_not_supplement

It is the synthetic Euclid I.17 fact that an interior angle of a triangle cannot be congruent to the corresponding exterior supplement.

Hence the same-ray secant branch is impossible.

The proposition-specific private helper is

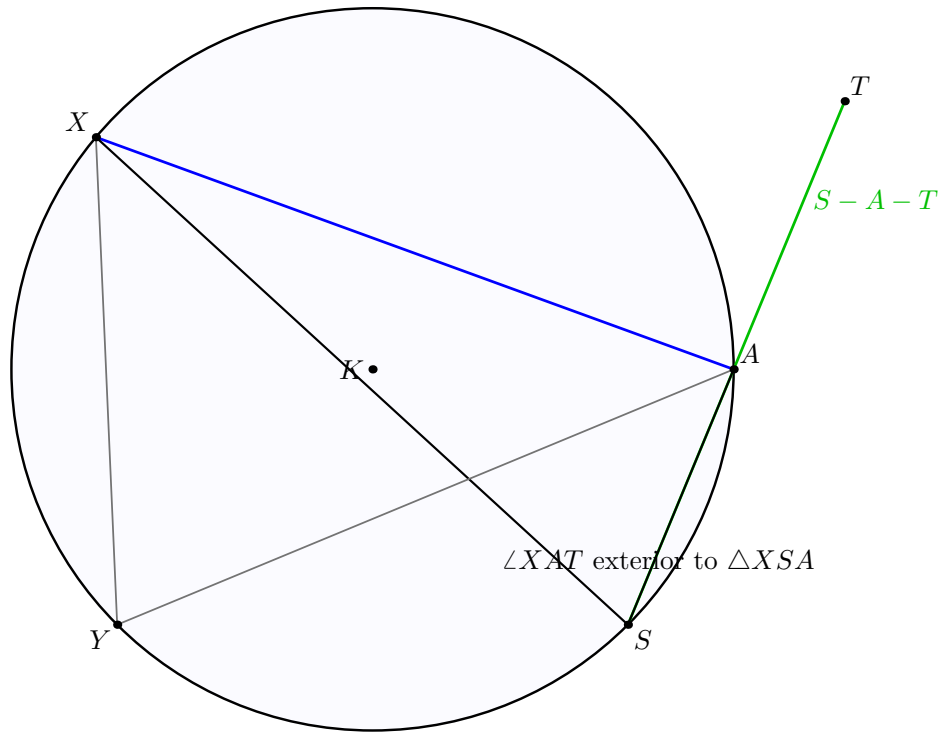
hilbert_IV17_sameRay_secant_impossible_stage57

10 Opposite-Ray Secant Branch

Assume now

$$S - A - T.$$

S, T opposite sides of AX



S, Y same side of AX

Figure 3: The impossible opposite-ray secant branch. Here $S - A - T$. Thus S, T are on opposite sides of the chord AX . Since T, Y are also opposite, S, Y lie on the same side of the chord.

Because A lies on the chord AX , the betweenness

$$S - A - T$$

shows directly that S, T lie on opposite sides of the chord.

The theorem hypothesis already gives

$$T \mid Y.$$

Thus T is opposite both S and Y , and the plane-separation theorem “two opposites are same side” gives

$$S \sim Y.$$

Apply the same-side half of IV.16 to X, A, S, Y :

$$\angle XSA \cong \angle XYA.$$

Using the hypothesis

$$\angle XAT \cong \angle XYA$$

and symmetry/transitivity gives

$$\angle XSA \cong \angle XAT.$$

But when

$$S - A - T,$$

the angle $\angle XAT$ is an exterior angle of triangle XSA .

The general exterior-angle theorem says that it cannot be congruent to the remote interior angle $\angle XSA$.

This contradiction eliminates the opposite-ray secant branch.

The proposition-specific helper is

```
hilbert_IV17_oppositeRay_secant_impossible_stage58
```

11 The Two Contradictions Compared

The two branches use different halves of IV.16 and different final contradictions.

Secant order	IV.16 input	Final contradiction
S, T same ray from A	S, Y opposite sides: IV.16 opposite-side	interior angle becomes congruent to its supplementary exterior angle
$S - A - T$	S, Y same side: IV.16 same-side	remote interior angle becomes congruent to the exterior angle

This is not an implementation accident. It is exactly the geometry of Forder’s proof.

12 Completion of the Tangent Converse

The top-level proof is now short.

Apply `hilbert_circle_line_second_or_tangent` to the line AT .

If the tangent alternative is returned, finish.

If a second circle point S is returned:

- if $S = T$, treat it as the same-ray case;
- otherwise apply `hilbert_secant_ray_order_split`;
- the same-ray result contradicts the first branch theorem;
- the opposite-ray result yields $S - A - T$ and contradicts the second.

Therefore the secant branch is impossible, and

AT is tangent at A .

The private complete theorem is

```
hilbert_forder_IV17_tangent_converse_stage59
```

and the production wrapper exports it as

```
hilbert_IV17_tangent_converse
```

13 Lean Formalization

The public surface is:

```
HilbertForderIV17TangentConverse  
hilbert_IV17_tangent_converse
```

The proposition-specific private proof layer contains:

```
hilbert_IV17_sameRay_secant_impossible_stage57  
hilbert_IV17_oppositeRay_secant_impossible_stage58  
hilbert_forder_IV17_tangent_converse_stage59
```

The former stage-56 triangle lemma is no longer conceptually owned by IV.17. Its production form is reusable interface infrastructure:

```
hilbert_triangle_angle_not_supplement
```

Likewise the secant-order split used by the top-level proof is the interface theorem

```
hilbert_secant_ray_order_split
```

The general circle-line dichotomy is

```
hilbert_circle_line_second_or_tangent
```

Thus the production module contains only the contradictions that are specific to IV.17 and the final assembly.

14 Relation to IV.16

IV.17 depends directly on both halves of IV.16.

The same-ray secant branch requires:

$$S \mid Y,$$

so it invokes

IV.16 opposite-side.

The opposite-ray secant branch requires:

$$S \sim Y,$$

so it invokes

IV.16 same-side.

This makes IV.17 the first theorem in the tangent block where the split public API of IV.16 pays off directly.

Had IV.16 been packaged only as a single theorem containing a disjunction, both contradiction lemmas would have had to reopen the chord-side classification. The present API avoids that duplication.

15 Hilbert and Book Zero Components

15.1 Circle-line incidence

The secant/tangent dichotomy is the global structural theorem behind IV.17. It says that no third geometric possibility exists for a line through the circle point A .

15.2 Ray order

The distinction

$$\text{SameRay}(A, S, T) \quad \text{versus} \quad S - A - T$$

is the decisive order classification.

The theorem `hilbert_secant_ray_order_split` is therefore not merely a technical convenience; it is the formal version of the source proof's implicit line-order case split.

15.3 Plane separation

Same-ray points on a carrier through the chord point lie on the same side of the chord. Opposite-ray points lie on opposite sides.

These relations are then combined with the hypothesis $T \mid Y$ to determine whether S, Y are same-side or opposite-side.

15.4 Book Zero supplements

The same-ray branch depends on transport of supplements of congruent proper angles. This replaces the source-level statement that two supplementary angles have equal supplements.

15.5 Euclid I.17 and exterior-angle order

The final contradiction in both branches is an angle-order statement for a proper triangle: an exterior angle is strictly larger than the relevant interior angle.

No angle measure is introduced; the contradiction is purely synthetic.

16 Foundational Boundary

IV.17 introduces no new tangent axiom.

Tangency is obtained from the previously proved circle-line dichotomy after the secant alternative has been eliminated.

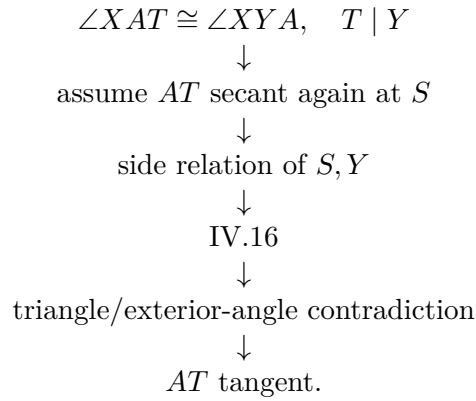
The proof uses:

- circle incidence;
- line and ray order;
- plane separation;
- IV.16 same-side and opposite-side;
- Book Zero supplement transport;
- Euclid I.17 / exterior-angle comparison.

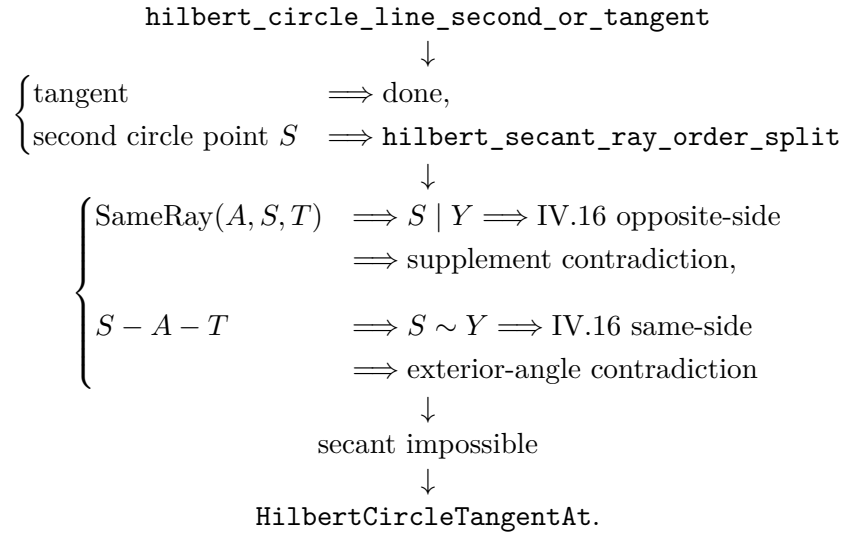
The theorem therefore reconstructs the converse tangent–chord criterion inside the existing Hilbert Euclidean plane theory.

17 Classical and Formal Dependency Structures

Forster's proof can be summarized as



The formal proof makes the line order explicit:



At module level:

```

HilbertInterfaceIV
Proposition4_16
  -> Proposition4_17

```

There is no production dependency on IV.19. The old stage-labelled secant-order helper was promoted to the interface precisely to preserve this acyclic direction.

18 Diagrammatic Audit

Three figures are useful.

1. The source tangent-converse configuration.
2. The same-ray secant contradiction.

3. The opposite-ray secant contradiction.

A separate tangent-only figure is unnecessary: the source figure already shows the intended conclusion, while the logical content of the proof lies in excluding the two secant orders.

19 Conclusion

Forder IV.17 is a particularly clean example of proof by geometric exhaustion.

The circle-line dichotomy says that the candidate carrier AT is either tangent or secant. In the secant case the second intersection S is classified relative to the ray AT .

If S lies on the same ray, IV.16 opposite-side makes the angle at S a supplement of the prescribed inscribed angle, and Book Zero transport forces an impossible triangle-angle supplement.

If S lies on the opposite ray, IV.16 same-side identifies the angle at S directly with the prescribed inscribed angle, making it congruent to an exterior angle of triangle $XS A$.

Both secant orders are impossible. Tangency is therefore forced.

The formal proof also illustrates the value of the Book IV refactor: general secant-order and triangle-angle lemmas live in `HilbertInterfaceIV`, while `Proposition4_17.lean` owns only the two proposition-specific contradictions and the final tangent-converse assembly.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.